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Do Big-Brained Animals Play More? Comparative Analyses of Play and Relative Brain Size in Mammals

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It has been hypothesized that play is more likely to be present in larger brained species. We tested this hypothesis in mammals using independent contrasts, a method that controls for phylogenetic relatedness. Comparisons across 15 orders revealed that the prevalence and complexity of play was significantly correlated with brain size, with larger brained orders having more playful species. Three orders, Rodentia, Marsupialia, and Primates, were used for within-order comparisons among species and, where possible, among families. The comparisons were not significant for rodents or primates, and those for marsupials yielded inconsistent results. Therefore, although a strong relationship is present at the highest taxonomic level of comparison, it diminishes or evaporates at lower level comparisons.

Following an extensive survey of the comparative distribution of play, Fagen (1981) concluded that play is a relatively rare behavior, with unambiguous cases found only in tetrapod vertebrates, especially birds and mammals. Fagen also noted that the distribution of play among vertebrates closely mirrors the evolution of brain size. Amphibians have smaller relative brain sizes than reptiles, and reptiles have smaller brains than birds, which in turn have smaller brains than mammals (Jerison, 1973). With regard to play, no convincing cases of playlike phenomena have been documented for amphibians, and only a couple of cases have been established for reptiles (Burghardt, 1998a; Burghardt, Ward, Roscoe, 1996; Kramer & Burghardt, 1998).

Play in birds can be as elaborate as that seen among mammals, but judging from descriptions of such complex play, it appears limited to a few corvids (Heinrich & Smolker, 1998; Pellis, 1981) and parrots (Diamond & Bond, 1999; Skeate, 1985). For the majority of birds, play is either absent (Fagen, 1981; Ficken, 1977)

or rudimentary (e.g., Blumstein, 1990; Sherman, 1995; Watson, 1993). Play appears to be most evident in the bird groups with the largest brains (Ortega & Bekoff, 1987). Similarly, although overall play is more prevalent in mammals than in birds, those mammalian orders in which play is most prevalent and most complex are those such as carnivores and primates (Fagen, 1981), which are also those with the largest brains (Jerison, 1983).

The relationship between play and brain size has recently been tested quantitatively in Australian marsupials, and a significant, positive correlation was found (Byers, 1999). That is, larger brained animals are more likely to engage in play than are smaller brained ones. Other considerations further support the plausibility of such a relationship. More complex behavioral output is expected to require more neural tissue (Jerison, 1973). For example, larger brained birds appear to use more complex and innovative feeding techniques (Lefebvre et al., 1998; Lefebvre, Whittle, Lascaris, & Finklestein, 1997). Furthermore, the brain-size-to-behavior correlation appears to extend to specific brain areas associated with particular functions. This is illustrated by the finding that birds with more complex food-storing behavior have a larger hippocampal volume (Hampton, Sherry, Shettleworth, Khurgel, & Ivy, 1995). Similarly, given that more frequent and complex forms of play are often considered to reflect greater levels of sophistication in neural control mechanisms (Fagen, 1981; Parker & McKinney, 1999), it may be expected that such play should be coupled with greater brain size. Analyses of traits that are correlated with both brain size and play, such as the length of the juvenile phase (Joffe, 1997) and the degree of growth occurring postnatally (Pellis & Iwaniuk, 2000b), further reinforce this expectation. Finally, species in which play continues into adulthood appear to be the largest brained members of their taxa (Brueggeman, 1978). Indeed, a comparative analysis of the prevalence of play in adult primates showed that species that have more of their brain development

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occurring postnatally play more as adults (Pellis & Iwaniuk, 2000b).

Although the available data relating brain size and play are compelling, there are some difficulties with this issue. First, broad comparisons among vertebrate classes and orders are qualitative and do not statistically evaluate the effects of confounding variables (Fagen, 1981; Ortega & Bekoff, 1987). That is, the conclusions are derived from patterns of overlap in brain size and play, not an empirical evaluation of the relationship. Second, the one quantitative study available (Byers, 1999) has some shortcomings. The data on the play of Australian marsupials were averaged for families, and so outliers could have had unpredictably large effects on the values used, especially in cases where only one or two species were used to represent a family. In addition, the families used for the statistical analysis were treated as independent data points, which is not valid given the different degrees of phylogenetic relatedness among those families (Harvey & Pagel, 1991; Iwaniuk, Pellis, & Whishaw, 1999b). It should be noted, however, that to some degree using averages for larger taxonomic levels has been applied as a means of controlling for the effects of relatedness (Harvey & Pagel, 1991). In this article, the hypothesis that brain size predicts the prevalence and complexity of play was tested for mammals using modern statistical procedures that correct for phylogenetic relationships (Iwaniuk, Pellis, & Whishaw, 1999a, 1999c; Iwaniuk et al., 1999b; Pellis & Iwaniuk, 1999a, 1999b, 2000b).

The analysis was in two parts. In the first, interorder comparisons of play and brain size were conducted using all the mammalian orders for which such data could be obtained with reasonable confidence. In the second, intraorder comparisons were conducted within three mammalian orders, and they involved both family- and species-level comparisons. The following three criteria had to be met for selecting appropriate mammalian orders for the intraorder analyses: (a) Data had to be available for a range of species across several families. (b) The data on play across these species had to be sufficiently unambiguous so as to provide a meaningful complexity or prevalence rating. (c) The order had to have a reasonably well resolved phylogeny. Species from two orders, marsupials and rodents, were used to test the hypothesis that brain size predicts the distribution of play in juveniles. The third order, primates, was used to test whether relative brain size affects the retention of playfulness in adulthood.

Fagen (1981) also noted that "play and a highly developed cerebral cortex go together" (p. 219), suggesting that it may not be overall brain size that is important, but the relative size of the neocortex. This suggestion is plausible given that much of the difference in brain size across vertebrate classes (Jerison, 1973) and among orders within classes (Jerison, 1983) arises from differences in the expansion of the neocortex (Finlay & Darlington, 1995). Therefore, the hypothesis that the size of the neocortex predicts the occurrence of play was also tested. Data to evaluate this hypothesis were available for two of the orders. These two taxa were used to evaluate whether increased volume of the neocortex predicted the prevalence of play in juveniles (marsupials) or the retention of play in adults (primates).

Method

Data Sets

In the literature on nonhuman animals, three types of play are generally reported: locomotor play, object play, and social play (Burghardt, 1998a, 1998b). Most commonly, locomotor play involves aspects of antipredator behavior, object play involves aspects of predatory behavior and food handling, and social play involves aspects of agonistic and sexual behavior (Power, 2000). Given the variation in the biological distribution of types of play (Fagen, 1981) and the variation in the detail available on different types of play (Pellis & Pellis, 1998a; Power, 2000), different kinds of play-rating scores had to be developed for the interorder and the intraorder comparisons.

With regard to brain and neocortex size, two methods were used. In the intraorder comparisons, the values used were derived from the variation present within each order. The relative brain size for members of each group was calculated by regressing log-transformed brain size against log-transformed body size within each order. The residuals were then used as measures of relative brain size. This method is commonly used for the removal of allometric effects to create "body-size"-free brain size measurements (e.g., Iwaniuk et al., 1999a; Joffe & Dunbar, 1997). Relative size of the neocortex was based on the residuals derived from regression analysis of log-transformed neocortex size against log-transformed brain size. Alternatively, encephalization quotients (EQs) and neocortex progression indexes were used as measurements of relative brain and neocortex size. The EQ is calculated by dividing the observed brain weight of each species by the expected brain weight given a specific body size (Jerison, 1973). The neocortex progression index is similarly derived, but the expected neocortex size is that for a given brain weight, not body size. A version of the EQ method was used for the interorder analysis. The specific play-rating scales and the measures of brain and neocortex size used for each of the analyses are described for each of the comparisons made.

Interorder Comparisons

Comparing play quantitatively across large taxonomic groupings is difficult because the limited availability of descriptive reports in some groups may artificially bias the values assigned—that is, it is unclear what would constitute an unbiased assemblage of species. Therefore, for this analysis, data were summarized so as to provide scores for each order. To do so, two values were developed for each order, with each score representing the modal tendency for the orders. One was based on the relative frequency of play being reported within an order. For example, although many members of the rodent order engage in play, not all do, whereas all members of the order primates do so. A 3-level scale was used for the measure of prevalence of play within orders. If play is rarely or never reported in an order, then that order was assigned a 1. If play is present in all or most species, then that order was assigned a 3. Any order falling in between these extremes was assigned a 2.

The second measure evaluated the complexity of the play, but again was based on the relative frequency of its occurrence within the species of each order. Play having the complexity present in primates and carnivores, that is, sequences containing many behavioral elements, was used as the standard for evaluation. Orders in which most species engage in such complex play were assigned a 3, whereas orders in which few or no species do so were assigned a 1. Again, orders falling in between these extremes were assigned a 2. The overall value for each order was based on the average of the two constituent scores (Table 1). This approach was taken because it is difficult to meaningfully assign weightings for different features of play across such large taxonomic comparisons. For example, all species of tree shrews appear to engage in social play and hence would score high for prevalence. However, the play is limited to chasing, rather than competitive physical contact (e.g., Sorenson, 1970), and so would score low for complexity. For both features of this rating system, any or all three types

Table 1
Encephalization Quotients (EQs) and Play Scores for 15
Mammalian Orders

Order	Play ^a	EQ ^b
Artiodactyla	2.0	0.775
Carnivora	3.0	1.257
Cetacea	3.0	2.071
Chiroptera	1.0	0.843
Edentata	2.0	0.866
Hyracoidea	2.5	0.906
Insectivora	1.0	0.681
Lagomorpha	1.5	0.592
Marsupialia	2.0	0.685
Monotremata	1.0	0.830
Perissodactyla	2.5	0.943
Primates	3.0	2.151
Proboscidea	3.0	1.591
Rodentia	2.0	1.011
Scandentia	2.0	1.369

^a Derived from descriptions provided in Fagen (1981). ^b Averaged from Appendix 6 in Eisenberg (1981).

of play were included, as their distribution varies across orders (Fagen, 1981; Power, 2000).

In many cases, it is not known which members of an order play and which do not, much less the relative differences in prevalence and complexity of the play that does occur. Therefore, it was not possible to select data on brain sizes that would reflect a fair sampling of all possibilities. Given these limitations, the EQ values for all the species listed in Appen-

dix 6 of Eisenberg (1981) were used to calculate an average EQ for each mammalian order (Table 1). These EQ values were used for comparison with the modal play ratings for each order.

Intraorder Comparisons

Two types of intraorder comparisons were conducted. First, the species comprising each order were directly compared to one another to determine whether variance in brain size predicted the distribution of play. Second, given that higher taxon averages have been used as a means for controlling for phylogenetic effects (Harvey & Pagel, 1991), it is possible that Byers' (1999) significant positive relationship reflected a family-level phenomenon. Therefore, for two of the orders (marsupials and primates), family averages for both the brain and the play values were compared. The rodents were not used because data on too few families were available. For the family-level comparisons, the family groupings, as shown in the appropriate tables below, were based on the taxonomy provided by McKenna and Bell (1997).

Rodents. For this study, play in this order was analyzed only for species in one suborder—the muroid rodents. Play in this taxon is primarily limited to locomotor and social play (Hole & Einon, 1984). A total of five different rating scales were used to score the play in this taxon: Three were related to social play and one to locomotor play. The fifth combined locomotor and social play to obtain an overall value for each species' play. Depending on the play ratings used, data were derived from the literature on 26 species from six different families (Table 2).

The complexity index was the most precise measure and was developed to compare play fighting, a major component of social play, within this suborder (Pellis & Iwaniuk, 1999b). Play fighting involves attack by one animal and defense by the partner, who may also launch a counterattack (Pellis, 1988). Seven criteria were incorporated to take these components

Table 2
The Five Play Scores and Relative Brain Sizes for the Species of Muroid Rodents Examined

Species	Complexity index	Play scores ^a				Brain size
		Wrestling	Play fighting	Locomotor play	Total play	
<i>Acomys cahirinus</i>	—	0	0	0	0	-0.057
<i>Apodemus flavicollis</i>	—	0	0	1	1	-0.007
<i>Cricetomys gambianus</i>	—	1	2	1	3	0.011
<i>Cricetus cricetus</i>	—	1	2	0	2	-0.145
<i>Meriones unguiculatus</i>	0.38	1	1	1	2	-0.033
<i>Mesocricetus auratus</i>	0.75	1	2	0	2	-0.151
<i>Micromys minutus</i>	—	1	1	0	1	0.008
<i>Microtus agrestis</i>	0.13	0	0	2	2	-0.113
<i>M. montanus</i>	0.31	1	1	0	1	-0.152
<i>M. ochrogaster</i>	0.50	1	2	0	2	-0.229
<i>M. pennsylvanicus</i>	—	1	1	0	1	-0.094
<i>Mus musculus</i>	0.13	0	0	2	2	-0.094
<i>Neotoma albigula</i>	0.13	0	0	2	2	0.114
<i>Notomys alexis</i>	0.13	0	0	0	0	0.134
<i>Ondatra zibethicus</i>	—	1	1	0	1	-0.012
<i>Onychomys leucogaster</i>	0.44	1	1	0	1	0.068
<i>Peromyscus gossypinus</i>	—	0	0	0	0	-0.003
<i>P. maniculatus</i>	0.38	1	1	0	1	-0.016
<i>Phodopus campbelli</i>	0.31	1	1	0	1	-0.182
<i>Pseudomys albicinctus</i>	0.00	0	0	0	0	0.041
<i>P. shortridgei</i>	0.00	0	0	0	0	0.090
<i>Rattus norvegicus</i>	0.94	1	2	2	4	-0.104
<i>R. rattus</i>	—	1	2	2	4	0.045

Note. Dashes indicate that those data were not available for that species.

^a The primary references for the various species can be found in Fagen (1981), Pellis and Iwaniuk (1999b, 2000b), and Pellis and Pellis (1998b).

of play fighting into account. These included the relative frequency of launching playful attacks, the presence of playful defense, the proportion of defensive tactics that led to the avoidance of bodily contact, the proportion of defensive tactics that promoted the maintenance of physical contact, the relative proportion of counterattacks directed at playful rather than agonistic body targets, the target sites competed for during play, and the degree of resemblance of the playful encounters to their adult counterparts. Both the criteria and how they were converted into a numerical score of play complexity are more fully described elsewhere (Pellis & Iwaniuk, 1999b; Pellis & Pellis, 1998a).

For most species reported in the literature, insufficient details of their play fighting are available, and so it was not possible to assign such species with a complexity score that would incorporate the variety of quantitative and qualitative features used above. Therefore, we developed two other rating scales so as to be able to incorporate a wider range of species. The first of these involved a 3-point scale: Those species for which no play fighting has been reported scored 0, and those in which the play fighting involves mutual wrestling scored a 2. Species for which play fighting is limited to brief contact, where the recipient avoids further contact rather than promotes wrestling (e.g., Pellis & Pasztor, 1999), were given a score of 1. The second of the more inclusive rating scales simply scored whether protracted playful wrestling was present (1) or absent (0) for each species.

Locomotor play has not been as intensively studied as play fighting, and so developing an index that incorporated a number of criteria was not possible for this form of play. Therefore, locomotor play was rated on a 3-point scale comparable to one of the measures used for play fighting, with those species for which locomotor play has not been reported scoring 0. Species performing sporadic and relatively isolated locomotor elements, such as jumps, runs, and body shakes, were rated with a score of 1. The most complex form of locomotor play involved descriptions of locomotor patterns being strung together into complex sequences of movement (e.g., Pellis & Pellis, 1983; van Oortmerssen, 1971)—such species scored a 2. Finally, because some species have more play fighting than locomotor play, whereas others have more locomotor play than play fighting, the 3-point rating scales for these two types of play were summed together to provide an overall play score for each species (Table 2).

So as to derive relative brain size data that reflected the variation present in muroid rodents reasonably well, the data for the present analyses were derived from a comparison of 115 species, which encompassed animals from the entire geographic range of this taxon (i.e., Africa, Eurasia, Australasia, and the Americas). Such an approach made it less likely that the brain measurements were biased by the few species for which play has been described. As brain size for *Microtus agrestis* was not available from direct measurements, it was estimated by calculating the brain-body size regression for nine *Microtus* species ($\log_{10}\text{brain size} = 0.426\log_{10}\text{body size} - 0.946$). The average weight of *M. agrestis* (30g; taken from Eisenberg, 1981) was entered into this equation, and the brain size estimate was entered into the calculation of the overall muroid rodent regression line. The residual of *M. agrestis* from this line was then used in the present analysis. (It should be noted, however, that the relationships reported below were the same regardless of whether the *M. agrestis* brain size estimate was included or not).

Marsupials. Marsupials range widely in the content of their play, with some exhibiting all three forms of play and others emphasizing only one of the three forms (Fagen, 1981). Given that most reports are relatively sketchy with regard to the descriptive detail provided, we used Byers' (1999) 3-point rating scale. Play absent (1) included species for which play has not been observed or reported. Rudimentary play (2) included species where the play involves brief (<5 s) bouts, and the behavioral elements are few and relatively isolated in performance. Complex play (3) included species for which play is reported as being common, occurring in long bouts (~20 s), with many behavioral elements strung together (see Byers, 1999, for further details). A narrower rating scale was used for scoring all the species for the presence of playful wrestling. As was the case for the

rodents, each species was simply scored for whether protracted playful wrestling was present (1) or absent (0). The data set used in this article included more species in each of the families of Australian marsupials than used by Byers and also included data on New World marsupials, which were not included in Byers. The EQ values and neocortex progression indexes were derived from the archival collection of one of the authors (Nelson, 2000; Table 3).

Primates. Most instances of play in adult primates involve social play (Fagen, 1981). Therefore, the literature on 64 species of primates, which included representatives from all families, was evaluated so as to rate each species for its expression of adult social play (Table 4). Although adults play with juveniles and infants, such play varies markedly with partner availability to the young animals and is most often initiated by the younger animal (Pellis & Iwaniuk, 1999a). The most unambiguous measurement of the adult propensity to play is when adults engage other adults in this activity. Adult-adult play can involve chasing, approach and withdrawal, or prolonged wrestling, but as reports varied in the descriptive details provided, it was rarely possible to make species comparisons based on the relative performance of the different aspects of play. Therefore, species were rated for the prevalence of play rather than for the content.

The rating used for scoring the level of play in species was to assign a 0 for those species where such activity has not been observed, or if it has, is rare. Some researchers explicitly state that adult-adult play was not observed (Kinzey, Rosenberger, Heisler, Prowse, & Trilling, 1977). In other species, adult-adult play is so infrequent (e.g., rhesus macaque, *Macaca mulatta*) that it is either not noticed or reported in most studies (Symons, 1978), although the occasional study that involves exceptionally long periods of observation may include a report of a low frequency of such play (Brueggeman, 1978). In contrast, studies may yield high numbers of instances of adult-adult play in a fraction of that sampling period (e.g., stump-tailed macaque, *M. arctoides*; Rhine, 1973). Reports of adult-adult play for the same species from more than one researcher observing the subjects for samples of average length were given a rating above 0. Two contexts for adult-adult play are reported: one related to sexual behavior (i.e., courtship or pair formation) and involving heterosexual partners, and the other involving isosexual or heterosexual partners that interacted in clearly nonsexual contexts (see Pellis & Iwaniuk, 1999a, for a partial review). Each of these two contexts of play was scored separately for each species. A score of 2 was given to species for which play was reported as common, and a score of 1 to species where play was reported as relatively infrequent. A total score for each species was derived from adding the scores for play occurring in both sexual and nonsexual contexts. Data on relative brain size and neocortex size were obtained from various sources (Barton, 1999; Stephan, Frahm, & Baron, 1981; Swindler, 1998; see Table 4).

Comparative Analysis

Because both relative brain size (Pagel & Harvey, 1988) and play (Pellis & Iwaniuk, 1999a, 1999b) can be significantly affected by phylogenetic relatedness, we used independent contrasts (Felsenstein, 1985), a method that controls for relatedness, to evaluate the relationship between these variables. This method was chosen because it is the most statistically robust and powerful comparative method currently available (Diaz-Uriarte & Garland, 1996, 1998; Garland, Midford, & Ives, 1999).

Independent contrasts is a method that calculates contrasts of a trait at each node of a phylogeny and then scales them according to an expected standard deviation derived from branch lengths. Because branch lengths were unknown for parts of the primate (Purvis, 1995) and marsupial (Kirsch, Lapointe, & Springer, 1997) phylogenies and for the entire rodent phylogeny, equal branch lengths were assumed for all the comparisons within the three orders. For the interorder comparison, three different branch-length models were used: equal, unequal but arbitrary (Grafen, 1989), and actual (derived from Novacek, 1993). Contrasts for each com-

Table 3
The Play Scores and Relative Brain (EQ) and Neocortex Sizes for the Marsupials Examined

Family ^a	Species	Play frequency ^b	Play wrestling	EQ	Neocortex
Acrobatidae	<i>Acrobates pygmaeus</i>	1	—	192	507
Burramyidae	<i>Burramys parvus</i>	1	—	181	420
	<i>Cercartetus nanus</i>	1	—	213	515
Dasyuridae	<i>Antechinomys laniger</i>	1	0	208	397
	<i>Antechinus stuartii</i>	1	0	196	430
	<i>Dasyercus cristicaudata</i>	3	1	215	443
	<i>Dasyuroides byrnei</i>	3	1	181	308
	<i>Dasyurus hallucatus</i>	3	1	149	312
	<i>Dasyurus maculatus</i>	3	1	170	391
	<i>Dasyurus viverrinus</i>	3	1	165	355
	<i>Phascogale tapoatafa</i>	1	0	181	343
	<i>Planigale ingrami</i>	1	0	84	106
	<i>Planigale maculata</i>	1	0	129	203
	<i>Sarcophilus harrisii</i>	3	1	152	324
	<i>Sminthopsis crassicaudata</i>	2	0	185	349
Didelphidae	<i>Didelphis virginianus</i>	1	0	134	239
	<i>Monodelphis domestica</i>	2/3 ^c	1	144	237
Macropodidae	<i>Aepyprymnus rufescens</i>	2/3 ^c	1	216	677
	<i>Bettongia penicillata</i>	3	0	273	824
	<i>Dendrolagus dorianus</i>	3	1	228	—
	<i>Dendrolagus lumholtzi</i>	2/3 ^c	1	205	654
	<i>Dorcopsis luctuosa</i>	2/3 ^c	1	243	727
	<i>Hypsiprymnodon moschatus</i>	2	0	263	736
	<i>Lagorchestes conspicillatus</i>	2	1	180	567
	<i>Macropus agilis</i>	3	1	165	519
	<i>Macropus antilopinus</i>	3	1	160	496
	<i>Macropus dorsalis</i>	3	1	203	646
	<i>Macropus eugenii</i>	3	1	227	828
	<i>Macropus fuliginosus</i>	3	1	198	—
	<i>Macropus giganteus</i>	3	1	176	630
	<i>Macropus parma</i>	3	1	232	788
	<i>Macropus parryi</i>	3	1	180	568
	<i>Macropus robustus</i>	3	1	174	606
	<i>Macropus rufogriseus</i>	3	1	185	592
	<i>Macropus rufus</i>	3	1	169	541
	<i>Petrogale assimilis</i>	3	1	194	581
	<i>Petrogale lateralis</i>	3	1	202	664
	<i>Petrogale penicillata</i>	3	1	231	742
	<i>Potorous tridactylus</i>	2/3 ^c	0	279	726
	<i>Thylogale billardieri</i>	3	1	196	593
	<i>Thylogale stigmatica</i>	3	1	211	655
	<i>Thylogale thetis</i>	3	1	181	565
	<i>Wallabia bicolor</i>	3	1	185	564
Myrmecobiidae	<i>Myrmecobius fasciatus</i>	3	0	211	—
Peramelidae	<i>Isodon obesulus</i>	1	—	134	192
	<i>Perameles gunii</i>	1	0	170	245
	<i>Macrotis lagotis</i>	2	0	207	—
Petauridae	<i>Dactylopsila trivirgata</i>	1	0	320	1120
	<i>Gymnobelideus leadbeateri</i>	2	1	306	976
	<i>Petauroides volans</i>	1	—	112	319
	<i>Petaurus australis</i>	3	1	275	914
	<i>Petaurus breviceps</i>	3	1	262	827
	<i>Pseudocheirus peregrinus</i>	2	—	176	504
Phalangeridae	<i>Trichosurus vulpecula</i>	2	—	205	586
Phascolarctidae	<i>Phascolarctos cinereus</i>	2	1	134	364
Tarsipedidae	<i>Tarsipes rostratus</i>	2	—	175	523
Vombatidae	<i>Lasiorninus latifrons</i>	3	1	164	520
	<i>Vombatus ursinus</i>	3	1	197	609

Note. EQ = encephalization quotient. Dashes indicate that those data were not available for that species.

^a Familial affiliation was taken from McKenna and Bell (1997). ^b Partly from "The Distribution of Play Behavior Among Australian Marsupials," by J. A. Byers, 1999, *Journal of Zoology, London*, 247, p. 352. Copyright 1999 by Cambridge University Press. Adapted with permission. Estimates for additional species are from Coulson (1989), Fadem and Corbett (1997), Hunsaker and Shupe (1977), Jarman (1991), McManus (1970), Miessner and Ganslosser (1985), Russell (1984), and Watson (1998). Various researchers (R. Goldingay, K. Handasyde, G. McKay, D. Lindenmayer, J. Seebeck, and P. Woolley) also provided us with unpublished observations that provided support for the play ratings used. ^c Species with a 2/3 score were treated as having either score for their play frequency.

Table 4
 Play Scores and Relative Brain and Neocortex Sizes for the Primates Examined

Family ^a	Species ^{b,c}	Sex play	Non-sex play	Total play	Brain size	Neocortex size
Atelidae	<i>Alouatta caraya</i>	0	2	2	-0.165	0.006
	<i>Aotus trivirgatus</i>	0	1	1	0.118	0.030
	<i>Ateles geoffroyi</i>	1	2	3	0.093	0.009
	<i>Brachyteles arachnoides</i>	0	0	0	0.049	—
	<i>Cacajao calvus</i>	—	2	—	0.150	—
	<i>Callicebus moloch</i>	0	0	0	0.065	0.030
	<i>Callicebus torquatus</i>	0	0	0	0.075	—
	<i>Cebus apella</i>	0	1	1	0.252	0.055
	<i>C. capucinus</i>	0	1	1	0.267	—
	<i>Lagothrix lagotricha</i>	0	2	2	0.107	0.009
	<i>Pithecia pithecia</i>	0	1	1	0.065	—
	<i>Saimiri sciureus</i>	1	1	2	0.236	0.063
Callitrichidae	<i>Callimico goeldii</i>	0	0	0	-0.034	0.053
	<i>Callithrix jacchus</i>	1	1	2	0.079	0.058
	<i>Cebuella pygmaea</i>	0	0	0	0.055	0.069
	<i>Leontopithecus rosalia</i>	0	0	0	0.063	—
	<i>Saguinus fuscicollis</i>	0	1	1	0.04	—
Cercopithecidae	<i>Cercocebus albigena</i>	0	0	0	0.054	0.014
	<i>Cercopithecus aethiops</i>	0	0	0	0.031	—
	<i>C. diana</i>	0	0	0	0.174	—
	<i>C. neglectus</i>	0	0	0	0.005	—
	<i>Colobus badius</i>	0	0	0	-0.103	0.020
	<i>C. guereza</i>	0	1	1	-0.128	—
	<i>Erythrocebus patas</i>	0	0	0	0.028	0.046
	<i>Macaca arctoides</i>	0	2	2	0.022	—
	<i>M. fuscata</i>	0	0	0	-0.026	—
	<i>M. mulatta</i>	0	0	0	0.113	0.032
	<i>M. nigra</i>	0	0	0	-0.014	—
	<i>M. radiata</i>	0	1	1	0.077	—
	<i>Mandrillus sphinx</i>	0	0	0	-0.018	—
	<i>Papio anubis</i>	0	0	0	-0.063	0.011
	<i>P. cynocephalus</i>	0	0	0	-0.007	—
	<i>P. hamadrayas</i>	0	0	0	-0.011	—
	<i>P. ursinus</i>	0	0	0	0.013	—
	<i>Presbytis entellus</i>	0	0	0	0.001	—
	<i>P. johnii</i>	0	0	0	-0.222	—
	<i>Pygathrix nemaeus</i>	1	1	2	0.003	0.007
	<i>Theropithecus gelada</i>	0	0	0	-0.081	—
Cheirogaleidae	<i>Microcebus murinus</i>	0	0	0	-0.058	-0.026
	<i>Phaner furcifer</i>	—	1	—	-0.086	—
Daubentoniidae	<i>Daubentonia madagascariensis</i>	1	0	1	0.055	-0.083
Hominidae	<i>Gorilla gorilla</i>	0	1	1	-0.189	-0.035
	<i>Homo sapiens</i>	1	2	3	0.427	-0.029
	<i>Hylobates lar</i>	0	1	1	0.188	0.004
	<i>H. syndactylus</i>	0	1	1	0.009	—
	<i>Pan paniscus</i>	1	2	3	-0.080	—
	<i>P. troglodytes</i>	1	2	3	0.104	-0.003
	<i>Pongo pygmaeus</i>	1	0	1	-0.024	—
	<i>Avahi laniger</i>	0	0	0	-0.316	-0.053
Indridae	<i>A. occidentalis</i>	0	0	0	-0.212	-0.05
	<i>Propithecus verreauxi</i>	0	1	1	-0.261	—
Lemuridae	<i>Eulemur fulvus</i>	0	1	1	-0.149	-0.028
	<i>E. mongoz</i>	0	0	0	-0.095	—
	<i>Lemur catta</i>	0	1	1	-0.155	—
	<i>Lepilemur mustelinus</i>	2	2	4	-0.112	—
	<i>Varecia variegata</i>	0	1	1	-0.153	-0.073
Loridae	<i>Galago sengalensis</i>	0	2	2	0.030	-0.035
	<i>Galagoides demidovii</i>	1	—	—	0.129	-0.003
	<i>Loris tardigradus</i>	2	2	4	0.030	0.031
	<i>Nycticebus coucang</i>	2	2	4	-0.105	-0.027
	<i>Otolemur crassicaudatus</i>	0	2	2	-0.232	-0.053
	<i>Perodicticus potto</i>	2	2	4	-0.078	-0.048
Tarsiidae	<i>Tarsius spectrum</i>	0	1	1	-0.112	—
	<i>T. syrichta</i>	1	0	1	0.081	—

Note. Dashes indicate that those data were not available for that species.

^a Familial affiliation taken from McKenna and Bell (1997). ^b Primary references for play for the various species can be found in Fagen (1981) and Pellis and Iwaniuk (1999a, 2000a, 2000b). ^c Neural data were derived from Barton (1999), Stephan et al. (1981), and Swindler (1998).

parison were calculated with the CAIC computer program (Purvis & Rambaut, 1995).

Prior to regression analyses, the contrasts were checked for proper standardization (Garland, Dickerman, Janis, & Jones, 1993). For most of the comparisons, the equal branch-length model properly standardized the contrasts, as determined by regressing the absolute value of the contrasts against their standard deviations (Garland, Harvey, & Ives, 1992). The comparisons that were not properly standardized were those between the EQs and play scores for marsupials and the comparison across orders. For the marsupial comparisons, application of Grafen's (1989) method of arbitrary (i.e., unequal) branch lengths also did not properly standardize the data. We therefore transformed the unequal branch lengths by using the square root of the values (as suggested by Garland et al., 1992) to obtain properly standardized contrasts.

For interorder comparisons, the EQ values were not properly standardized by any of the following: square-root, cube-root, or log transformations of the branch lengths derived from the different models, or of the original data (shown in Table 1). Because improperly standardized contrasts can result in inflated Type I error rates (Garland et al., 1992), we analyzed the interorder data in two different ways. First, we performed a regression analysis using orders as independent data points. Although this violates the assumptions of independence of the data because of a shared ancestry, the use of higher taxon averages has been used in other studies as a means for controlling for phylogenetic effects (Harvey & Pagel, 1991). Second, we analyzed the contrasts of the equal branch-lengths model despite the improper standardization of the EQ contrasts. We chose this model because it tends to be more conservative than the other branch-length models (Martins & Garland, 1991), and its EQ contrasts were the least compromised by the improper standardization: EQ-branch-length regression, $F(1, 12) = 7.32, p = .02$.

Once standardized, the contrasts were checked for normality using Shapiro-Wilk tests (JMP-IN, 1997). All the brain size contrasts were not normally distributed, and therefore square-root, and in some cases cube-root, transformations were applied. Many of the play scores were also non-normally distributed, but given their range of positive and negative values, simple transformations were not possible. None of the family-level comparisons were successfully transformed, and so raw scores were used. Similarly, because of the differing procedures needed to transform different data sets (e.g., square root for rodent locomotor play, cube root for primate sex play), all the calculations for the species-level comparisons were repeated using the raw, untransformed data. The contrasts were then regressed against one another through the origin (Garland et al., 1992), using least-squares linear regression. It should be noted that we also performed Spearman rank correlations on all non-normal contrasts, and these corroborated the results reported below.

In order to use the independent contrasts method, a known phylogeny is required. The phylogenetic relationships among the mammalian orders used was derived from that provided by Liu and Miyamoto (1999). For primates, we used the phylogeny provided by Purvis (1995). The phylogeny for muroid rodents is based on a variety of studies (Brownell, 1983; Catzefflis, Dickerman, Michaux, & Kirsch, 1993; Chaline & Graf, 1988; Watts & Baverstock, 1995) and is presented in graphical form elsewhere (Pellis & Iwaniuk, 2000b). The phylogeny for marsupials was also based on a variety of studies and is shown in Figure 1.

Results

Analysis of the raw EQ values and play scores across mammalian orders indicated that there was a significant, positive relationship, $F(1, 13) = 14.80, p < .01, r^2 = .50$ (Figure 2a). This relationship was moderately improved on by using a polynomial regression model, indicating that there may be a stepwise, rather than a monotonic, relationship between play and brain size, $F(2, 12) = 8.20, p < .01, r^2 = .58$ (Figure 2b). Similarly, comparison

of the unstandardized independent contrasts also revealed a significant polynomial relationship, $F(2, 11) = 19.57, p < .01, r^2 = .77$. These results suggest that throughout mammalian evolution, increases in the prevalence and complexity of play have been correlated with increases in EQ.

For the intraorder familial comparisons, only 1 of the 10 analyses revealed a significant relationship, that between sex play and brain size in primates (Table 5). However, even this relationship disappeared once humans, which were extreme outliers for the family Hominidae (see Table 4), were removed. Although none of the species-level intraorder comparisons involving transformed scores were significant (Table 6), three of the comparisons involving untransformed scores were significant. For primates, brain size and sex play were significantly correlated, $F(1, 58) = 9.09, p < .01, r^2 = .14$, but again, as for the family-level comparison, this appeared to be due to the inclusion of humans. For marsupials, both EQ and play frequency, $F(1, 50) = 7.57, p = .01, r^2 = .13$, and EQ and play wrestling, $F(1, 42) = 4.82, p = .03, r^2 = .10$, were significantly correlated. Therefore, it appears that at the level of family and species differences within orders, the relative size of the brain or the neocortex does not predict the prevalence or complexity of play in either juvenile rodents or adult primates. Some evidence supports such a relationship for marsupials, but even for this group, the evidence is far from consistent.

Discussion

Do You Need a Large Brain to Play?

Comparison across mammalian orders revealed a significant positive relationship between play and brain size. That is, orders of mammals containing larger brained species also contain more species that are playful. This finding thus provides quantitative support for the hypothesis that play is more prevalent and more complex in larger brained species and is consistent with the qualitative trends shown for tetrapod vertebrate classes (Fagen, 1981) and avian orders (Ortega & Bekoff, 1987). However, the intraorder comparisons involving families and species did not yield a consistent relationship between brain size and the expression of play.

None of the comparisons for juvenile muroid rodents revealed significant correlations for play and overall brain size. Similarly, play in adult primates was not correlated to either overall brain size or neocortex size. The one significant relationship in primates, that between sex play and overall brain size, evaporated once humans, who are extreme outliers for the group, were removed. For the marsupials, the findings were mixed. Our results for the comparisons of brain size and play in marsupials do not replicate the findings of Byers (1999), who did find a significant positive correlation at the family level. However, unlike Byers' study, our analyses incorporated a method for accounting for the effects of relatedness (independent contrasts), and the data set was greatly expanded, not only to include more species per family of Australian marsupials, but also to include species from the Americas. None of the family-level comparisons in the present study were significant. For the species-level comparisons, when using untransformed data, both the frequency of play and the occurrence of playful wrestling were significantly correlated with overall brain size. This was not the case when the transformed data were used, nor were any of the comparisons with the neocortex significant,

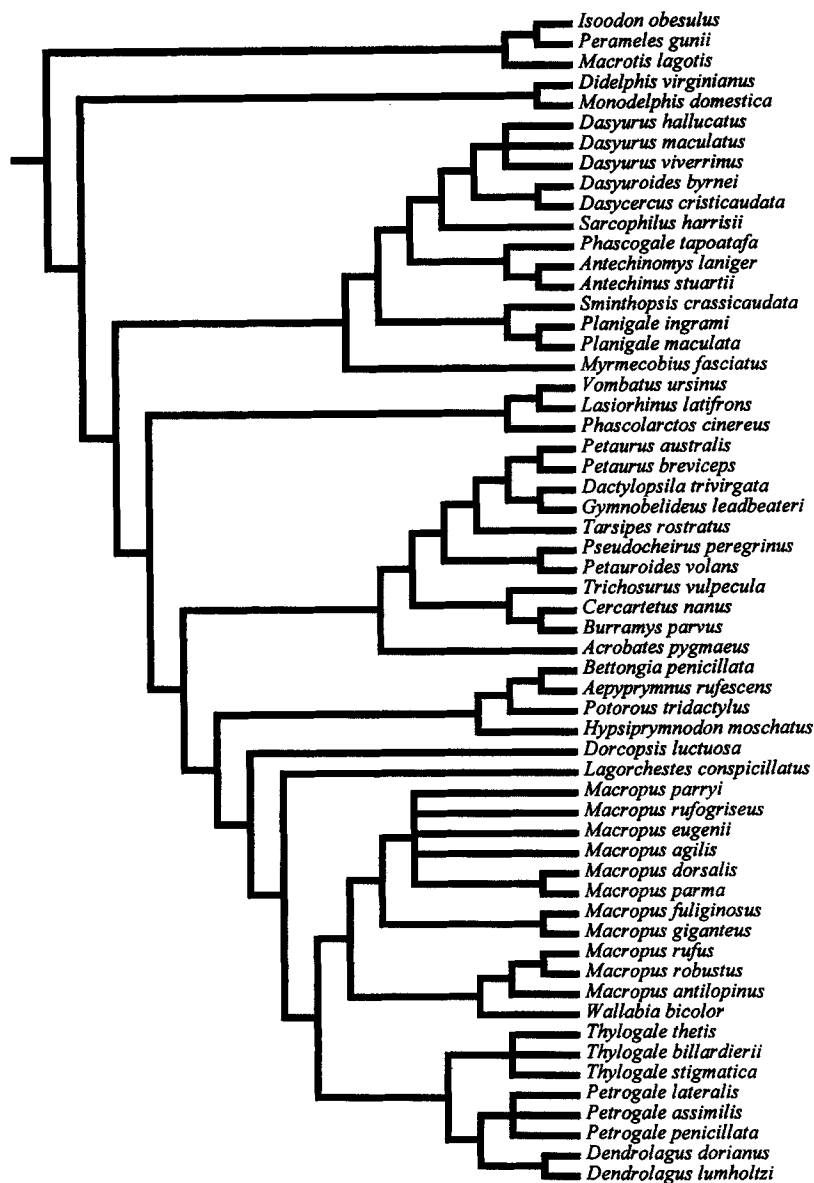


Figure 1. The phylogenetic relationship of the species of marsupials used in this study. The phylogeny was derived from a number of sources (Burk, Westerman, & Springer, 1998; Flannery, 1989; Kirsch et al., 1997).

although at the family level, the correlation between neocortex size and play frequency approached significance ($p = .06$).

Therefore, the present analyses provide conflicting evidence for the hypothesis that play is more prevalent or more complex in large-brained species: This relationship is robust for comparisons between orders but either absent or weak in within-order comparisons. Indeed, it should be noted that for all three orders, some of the most playful species had the smallest relative brain sizes, and some of the largest brained species were the least playful. These findings suggest that only large increases in brain size, as evident across orders, may have had a role in the evolution of play within the class Mammalia. This is consistent with other comparisons of large taxonomic units (i.e., classes, orders; see Fagen, 1981; Ortega & Bekoff, 1987). The present findings, however, also

suggest that smaller differences in brain size among species within orders are not important for evolutionary changes in play. The negligible role of within-order differences is consistent with the effects of domestication and with experimental ablations of the neocortex.

Domesticated breeds are more playful than their wild progenitors (Budiansky, 1999), yet they exhibit reductions in brain size of up to 30% (Kruska, 1988). Indeed, it is the brain areas that develop last, such as the neocortex, that exhibit the largest reductions in size (Kruska, 1988). This is consistent with our analyses that show that neither overall brain size nor neocortex size predict play prevalence or complexity in most cases. Similarly, removal of the neocortex at birth does not diminish play in either juvenile hamsters (Murphy, MacLean, & Hamilton, 1981) or juvenile rats

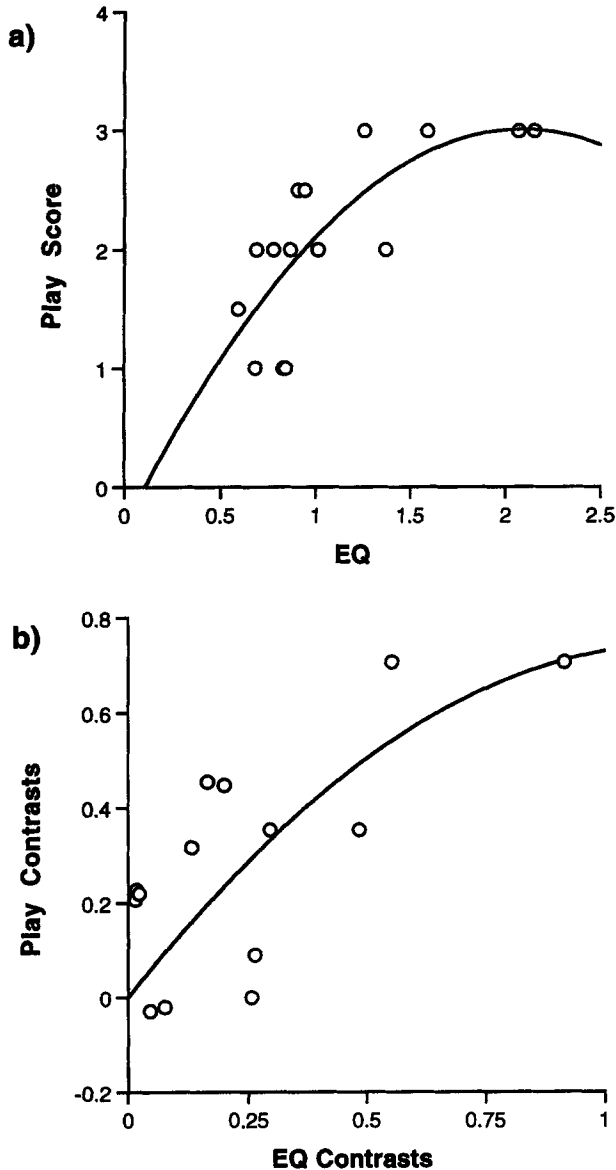


Figure 2. Scatter plots showing the relationship between brain size (i.e., encephalization quotient or EQ) and play among mammalian orders using raw scores (a) and independent contrasts (b). The line of best fit for both (a) and (b) was derived from a polynomial model, as the scores were not transformed to logarithmic values.

(Panksepp, Normansell, Cox, & Sivi, 1994; Pellis, Pellis & Whishaw, 1992), further showing that the cortex need not be physically present for play to occur. The significant comparisons found in this study for marsupials may reflect the wide variation in brain sizes in this order compared with the others. Indeed, the only significant correlations found for primates involved the inclusion of humans, which again would have markedly increased the variance in the data set.

A model consistent with these findings is that the relationship between play and brain size may be stepwise, so that with an expansion of the brain above a threshold level, play prevalence and

Table 5

Results of the Regression Analysis of Family-Level Independent Contrasts of Play Scores Against Relative Brain and Neocortex Size for Marsupialia and Primates

Taxon and neural measurement	Type of play score	<i>F</i>	<i>df</i>	<i>r</i>	<i>p</i>
Marsupialia					
Brain	Play frequency	2.18	1, 10	.42	.17
	Play wrestling	0.94	1, 6	.37	.37
Neocortex	Play frequency	4.68	1, 9	.59	.06
	Play wrestling	1.96	1, 5	.53	.22
Primates					
Brain	Sex play	10.02	1, 8	.75	.01
	Sex play ^a	3.57	1, 8	.56	.10
	Non-sex play	0.11	1, 8	.12	.74
	Total play	1.57	1, 8	.41	.25
Neocortex	Sex play	2.95	1, 7	-.55	.13
	Non-sex play	0.50	1, 7	-.26	.50
	Total play	1.73	1, 7	-.45	.23

^a Excluding humans.

complexity increases. However, variation within the range between one threshold level and the next has no further bearing on play. At a broad level, this implies that the increase in brain size shown by the average carnivore compared with the average rodent increases the likelihood that play is present among more species of carnivores. However, which carnivores play, and how they do so, may have little to do with intraorder variation in brain size. Thus although a large increase in brain size increases the likelihood that play will occur, whether it occurs or not depends on the presence of additional selective factors (Bekoff & Byers, 1998; Pellis & Iwaniuk, 1999a, 1999b, 2000b). This possibility is supported by our analysis showing that the strength of the relationship between

Table 6

Results of the Species-Level Independent Contrasts Analysis of Play Scores and Relative Brain Size in Rodents, Marsupials, and Primates

Taxon and neural measurement	Type of play score	<i>F</i>	<i>df</i>	<i>r</i>	<i>p</i>
Rodents					
Brain	Complexity index	2.83	1, 11	-.45	.12
	Play wrestling	0.41	1, 17	-.15	.53
	Play fighting	1.22	1, 17	-.26	.28
	Locomotor play	0.01	1, 17	-.03	.92
	Total play	0.76	1, 17	-.21	.40
Marsupials					
Brain	Play frequency	3.40	1, 50	.26	.07
	Play wrestling	3.87	1, 42	-.29	.06
Neocortex	Play frequency	1.35	1, 46	.17	.25
	Play wrestling	1.88	1, 39	-.22	.18
Primates					
Brain	Sex play	3.29	1, 58	-.01	.96
	Non-sex play	0.05	1, 59	.03	.83
	Total play	2.00	1, 57	.18	.16
Neocortex	Sex play	0.21	1, 29	.01	.65
	Non-sex play	0.73	1, 28	-.03	.40
	Total play	0.20	1, 28	-.08	.66

Note. None of the results were significant.

play and brain size among orders was improved by using polynomial regression models; this indicates that a stepwise model is superior to a linear model in explaining the relationship. Finally, the comparisons between families within two of the orders show that the stepwise effect seen in comparisons between orders either disappears or is at least greatly diminished. This further supports the idea that if brain size changes affect the prevalence and complexity of play, the changes involved are likely to be large ones.

The threshold model for the relationship between play and brain size can account for the seemingly contradictory data presented in this paper. Each stepwise increase in brain size increases the likelihood that play occurs and that it is of a more complex form. If this is so, then brain-size-related differences in play prevalence and complexity are likely to be most obvious when large taxonomic groupings are compared, such as classes (Fagen, 1981) and orders (Ortega & Bekoff, 1987; this study). As shown in this article, such threshold effects tend to disappear with lower level comparisons involving families or species. The special status of humans with regard to play provides additional support for the threshold model.

Humans are extreme outliers for both brain size and play (Fagen, 1981; Passingham & Ettlinger, 1974), with the content of human play being qualitatively different from other mammals (Garvey, 1990). As noted in the Method section above, play in nonhuman animals is typically categorized as locomotor, object, or social. Although there may be some overlap in these categories, detailed analyses reveal that they tend to remain developmentally and motivationally distinct (Gomendio, 1988; Pellis, 1991; Pellis, Pasztor, Pellis, & Dewsbury, 2000). In contrast, human children can mix and match behavioral elements derived from social, locomotor, and object play: This makes it difficult to classify all forms of human play using the categories adopted for nonhuman play (Burghardt, 1998b; Power, 2000). For example, although humans engage in a species-typical version of play fighting that is comparable to that of nonhuman species (Pellegrini & Smith, 1998), humans also engage in a variety of rule-bound social games that can involve the use of objects (Garvey, 1990); these do not have analogous forms in nonhuman species, except in a rudimentary manner, in apes (Jensvold & Fouts, 1993; Parker, 1996).

Threshold Versus Nonthreshold Brain-Play Relationships

Although the data presented in this study are more supportive of a threshold effect on the influence of brain size on play than on a gradualistic relationship, it is still possible that a gradualistic relationship exists. The failure to detect either familial or species effects in this study may have been due to the crudeness of the measures of play prevalence and complexity used. For example, comparison of the complexity of operations and cognitions performed during play show that humans achieve a much higher level of sophistication than other primates, with apes performing at intermediate levels (Parker, 1996; Parker & Milbrath, 1994). These cognitive differences in play appear to parallel brain size differences (McKinney, 1998; Parker & McKinney, 1999). It is possible, therefore, that significant differences in the play of species whose brains vary in size may have been missed by the measures of play prevalence and complexity used in this and other studies (e.g., Byers, 1999). Some apparent

interorder differences in the cognitive content of play fighting serve to illustrate this point.

The Norway rat has a more complex form of play fighting than any other muroid rodent as yet studied (Pellis & Iwaniuk, 1999b), but even so, the form of the play appears highly stereotyped, in that the attack and defense is rigidly centered around the play target (Pellis & Pellis, 1987, 1997). In contrast, in primates such as gorillas, although play fighting also generally has attack and defense sequences involving a specific body target (Schaller, 1963), they can markedly deviate from this pattern by attacking novel targets or by adopting novel forms of attack and defense (Pellis & Pellis, 1998b). The difference between the play fighting of rats and gorillas appears to reflect a qualitative difference in cognitive sophistication, whereas the difference in the play complexity of rats and other rodents does not. Small quantitative differences in the frequency of attack, defense, and counterattack appear capable of accounting for most of the species differences in the overt play seen among muroid rodents (Pellis & Pellis, 1998a; Pellis, Pellis, & Dewsbury, 1989). In all these rodent species, however, the targets competed over and the tactics used to do so do not change from the species-typical pattern (Pellis et al., 1989). In this regard, the idiosyncratic changes in the targets and tactics present in gorillas represent a qualitative change in the content of the play.

Differences in the cognitive content of play as is evident in rats and gorillas may similarly occur in families and species within the same order. For example, it may be that the play fighting of species in smaller brained families is more like that of the rat, whereas that of species in larger brained families may be more like that of the gorilla. If this is so, given our limited knowledge of the content of play in most species, it may not be so surprising that only the effects of the larger brain size differences can be statistically identified—that is, the differences that are large enough to be detected by relatively crude measures of play prevalence and complexity. To add to the problem, comparisons involving crude volumetric differences in brain size may also be inadequate for the analysis of the effects of differences in the brain on the expression of play. Either some subcomponents of the brain may be more critical than others, or the factors regulating their development may be of central importance (Aboitiz, 1996; McKinney, 1998). For example, the proportion of the brain developed postnatally predicts 60% of the variance in the distribution of play in adult primates (Pellis & Iwaniuk, 2000b). Given that overall brain size or size of the neocortex does not do so (this study), these data suggest that the manner in which the brain is developed may be more critical for the expression of play.

Conclusion

From the analyses in this study, it may be concluded that at the level of species within orders, neither of the brain features hypothesized to predict play, overall brain size and neocortex size (Byers, 1999; Fagen, 1981; Ortega & Bekoff, 1987), does so, or does so consistently. This was shown to be the case within three divergent mammalian orders, suggesting that our results are likely to be generalizable to other mammalian orders, if not to other vertebrate taxa. The study also found convincing evidence for a significant interorder effect of brain size on play, although a decisive test of this relationship requires using individual species' values from

each order. However, the species included in such a test need to be selected in a manner that does not bias either the brain values or the play values representing each order. Given the inadequate state of our descriptive knowledge of play in most mammalian species, developing such an unbiased data set is not currently feasible.

If both the inter- and intraorder results are true, then this strongly supports a stepwise relationship between brain size and play—that is, large differences in brain size affect play, but small differences do not. Alternatively, it is possible that the smaller effects can only be detected by detailed measurements of the cognitive content of play. An adequate evaluation of threshold and gradualistic models of the influence of brain size on play will require a data set that provides information on the fine-grained cognitive differences in the content of play across a wide range of species. In this regard, it is encouraging that researchers have recently begun to focus on the cognitive content of play (Bekoff & Allen, 1998; Biben, 1998; Pellis & Pellis, 1998b; Thompson, 1998).

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